

Tutorial: Automate your SP-API Calls Using a Python SDK

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# Tutorial: Automate your SP-API Calls Using a Python SDK

Automate your SP-API calls with Login with Amazon (LWA) token exchange and authentication.

This tutorial provides you with all the required details to generate a Python SDK with Login with Amazon (LWA) token exchange and authentication, to build your application seamlessly. You will learn the prerequisites required to build the Python SDK and also view an example using the Selling Partner Orders API and the Swagger Code Generator.

You can use this SDK to integrate Amazon marketplace features into your applications, including accessing product information, managing orders, handling shipments, and more.

## Prerequisites

To complete this tutorial, you need:

- A hybrid or SP-API app in draft or published state
- Integrated development environment (IDE) software
- Python (version 3.6 or later)
- `swagger-codegen-cli-2.3` (or later) This tool is used to generate Python client code from the SP-API's Swagger definitions.

Before your application can connect to the Selling Partner API, you must register it, and it must be authorized by a selling partner. If you do not have a hybrid or SP-API app, follow the steps to [register as a developer](#), [register your application](#), and [Authorizing Selling Partner API applications](#). Then, return to this tutorial.

Next, set up your workspace for the tutorial.

## Step 1. Set up your workspace

1. On your local drive, create a directory for this project, name it `SPAPI_Python_SDK`, and navigate to the new directory.
2. Clone the [client repo](#).

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## 3. Download the following tools.

1. IDE software (this walkthrough uses [Visual Studio](#) IDE on Windows OS)
2. Python (version 3.6 or later). You can download this software from [python.org](#).

## 4. Run the following command in your terminal to download the Swagger Code Jar:

Bash

```
`wget
https://repo1.maven.org/maven2/io/swagger/swagger-
codegen-cli/2.4.13/swagger-codegen-cli-2.4.13.jar
-O swagger-codegen-cli.jar`
```

5. Copy `swagger-codegen-cli.jar` into your local directory `C:\\SPAPI_Python_SDK`.6. Run the following command in your terminal to install the Python `backoff` library in your environment:

Bash

```
pip install backoff
```

7. In GitHub, go to <https://github.com/amzn/selling-partner-api-models/tree/main/models> and run the following command to clone the `selling-partner-api-models` repository to your local directory `C:\\SPAPI_Python_SDK`.

Bash

```
git clone https://github.com/amzn/selling-partner-
api-models
```

Now that you have completed the required setup, the next step is to generate the Python SDK with the authentication and authorization classes downloaded to your local directory `C:\\SPAPI_Python_SDK`.

## Step 2. Generate a Python client from Swagger definitions

1. Locate the Swagger JSON file for the [SP-API model](#) you want to use (for example, Orders API) from your local directory `C:\\SPAPI_Python_SDK`.
2. Run the following command in your terminal to generate client code. Make sure to replace the paths and API model with your settings.

Bash

Special Authorization Types

&gt;

## INTEGRATION

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```
java -jar /[path_to_swagger_jar]/swagger-codegen-
cli.jar generate -l python -t
/[path_to_mustach_resources]/resources/ -D
packageName=swagger_client -o
/[path_to_client_folder]/client/[SP-API_NAME] -i
/[path_to_model_folder]/models/[SP-API_NAME]/SP-
API.json
```

Now that you have generated a Python client, you need to integrate the authentication model.

## Step 3. Integrate the authentication module

1. Locate the `auth` and `spapi` client code folders in `C:\SPAPI_Python_SDK`, the directory where the SDK was downloaded.
2. Update the paths in the following Python files: `spapiclient.py` and `LwaRequest.py`. Each file contains instructions on what to update.

With authentication set up, you're now ready to set up the Python SDK package.

## Step 4. Set up the Python SDK package

1. Navigate to `C:\SPAPI_Python_SDK` the directory where the SDK was generated.
2. Use the following code to create a `setup.py` file. This file is required for packaging your SDK. Make sure you replace information in the example with information for your package and dependencies.

Delete an Application From Your Developer Account

## CALLING THE SP-API

Configuration Details

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Call Structure

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Development Tools

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Performance Management

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Python

```
from setuptools import setup, find_packages

setup(
    name='SellingPartnerAPIAuthAndAuthPython', #
    Replace with your package's name
    version='1.0.0', # Replace with your
    package's version
    package_dir={'': 'src'}, # Replace 'src' as
    necessary
    packages=find_packages(where='src'),
    install_requires=[line.strip() for line in
    open("requirements.txt", "r")],
    description='A Python SDK for Amazon Selling
    Partner API',
    long_description=open('README.md').read(),
    long_description_content_type='text/markdown',
    url='TBD'
)
```

With the Python SDK set up, you're now ready to create an instance of the Orders API.

## SOLUTION PROVIDER PORTAL

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**Tutorial: Automate your SP-API Calls Using a Python SDK**

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What is the Selling Partner Appstore?

List Your App on the Selling Partner Appstore

## Step 5. Create an instance of the Orders API and invoke an operation

### Note

This is a restricted operation and requires a Restricted Data Token (RDT) for authorization. Some operations require restrictions because they contain customer PII data. These operations require passing an additional security review. For more information on RDT, refer to the [Tokens API](#) guide.

The following is an example of how to use the Python SDK with the Orders API to make a `getOrders` request. Update the code with your information and then run the code in your terminal.

Python

```
if __name__ == "__main__":

    from auth.credentials import SPAPIConfig
    config = SPAPIConfig(
        client_id="Your client-id",
        client_secret="Your client-secret",
        refresh_token="Your refresh-token",
        region="NA", # Possible values NA, EU, FE,
and SANDBOX
        scope = None # Required for
grant_type='client_credentials' ; Possible values
"sellingpartnerapi::notifications" and
"sellingpartnerapi::migration"
    )

    from spapi.spapiclient import SPAPIClient

    # Create the API Client
    print("Config and client initialized...")
    api_client = SPAPIClient(config)

    marketplace_ids = ["ATVPDKIKX0DER"]
    created_after = "2024-01-19T00:00:00"

    orders_api =
api_client.get_api_client('OrdersV0Api')
    orders_response =
orders_api.get_orders(marketplace_ids=marketplace_ids,
created_after=created_after)
    print("Orders API Response:")
    print(orders_response)
```

### Caution

Never commit this file to your version control system as it contains sensitive information. Ensure these LWA credentials are stored securely in an encrypted format.

A status code of 200 means the API call was successful.

# Step 6. Connect to the Selling Partner API using the generated Python SDK

Run the following commands in your terminal to build and install your SDK locally:

```
Bash
python3 setup.py sdist bdist_wheel
```

```
Bash
pip install dist/{YourPackageName}-1.0.0-py3-none-any.whl
```

Run the following test script in your terminal to test the Python SDK:

```
Bash
python test.py
```

A status code of 200 means the API call was successful.

Updated 18 days ago

[← Tutorial: Automate your SP-API Calls Using a JavaScript SDK for Node.js](#)

[Tutorial: Automate your SP-API Calls using a prebuilt C# SDK →](#)

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